Higher National Diploma In Engineering (Civil)
Fourth Year, First Semester Examination – 2019
CE 3224 - Environmental Engineering

Instructions for Candidates:	No. of questions : 06
Answer five questions only	No. of pages : 03
State any assumptions you made.	Time : 3 hours

Q1: Wastewater treatment process has become popular in almost all the nations in the world.

i. State four reasons for treating wastewater before discharging into water bodies (04 Marks)

ii. State four physical characteristics of wastewater (04 Marks)

iii. What are the functions of a grit chamber and explain the importance of it in a treatment plant (04 Marks)

iv. Explain the processes involved in primary treatment of a wastewater treatment plant. (04 marks)

v. Briefly explain two types of biological decomposition of waste. (04 marks)

(Total 20 Marks)

Q2. Biochemical Oxygen Demand (BOD_5) is an important parameter used in waste water treatment process.

a. What are the reasons for the consideration of BOD as an important parameter used in waste water treatment process? (04 marks)

b. "Ultimate BOD does not depend on temperature whereas BOD_5 depends on temperature" Explain this statement (04 Marks)

c. "After pre-treatment and primary treatment secondary treatment is depending on the BOD level that needs to be reduced". Explain this statement. (04 marks)

d. What are the sources with nitrogen and phosphorus compounds as substances to wastewater? (04 marks)

e. Briefly explain the negative impacts of nitrogen and phosphorus available in wastewater on environment. (04 marks)

(Total 20 marks)

Q3 "Air pollution is a major issue in the globe"

a. Name four hazardous air pollutants and their sources. (04 marks)

b. What do you understand by adsorption system in air pollution control and state two examples (04 marks)

c. Explain "particulate pollutants" with examples (04 marks)

d. Briefly define the term "Municipal Solid Waste, (MSW)" (04 marks)

e. Briefly explain the current situation of managing the solid waste in Sri Lanka. (04 marks)

(Total 20 Marks)

Q4. ABC Company has appointed you as a coordinator for the design process and getting approval for their proposed factory building in Negombo. Being an expert in Building construction, answer to the following in terms of Central Environmental Authority (CEA) procedures and LEED certification.

a. What would be your first approach regarding CEA procedures? (02 marks)

b. What do you understand by Environmental impact assessment? (03 marks)

c. What is an "Environmental Recommendation"? What type of projects requires this? (05 marks)

d. What do you understand by "LEED" and briefly explain the objectives LEED certification. (04 marks)

e. List any six components that you will propose to include in proposed factory building design in order to gather points for LEED certification. (06 marks)

(Total 20 Marks)

Q5. "Noise pollution is sometimes nuisance and sometimes causes for permanent hearing disorders"

a. What is the meaning of 'Noise" and State three effects of noise (05 marks)

b. State three methods to reduce "Noise leakage" in a building (03 marks)

c. Briefly discuss about the existing regulations on noise in Sri Lanka (06 marks)

d. You have been asked to purchase and install a generator to your ATI. Briefly explain in detail at least three factors that you would consider for the specification of generator and installation techniques based on your knowledge in noise pollution control. (06 marks)

(Total 20 marks)

Q6. "Sludge treatment and solid waste management are problematic areas to which better solutions to be seek"

a. Briefly explain four sludge treatment processes? (04 marks)

b. Briefly explain "composting process" (04 marks)

c. What are the objectives of composting? (02 marks)

d. Briefly explain environmental effects of land fill. (04 marks)

e. Design a solid waste management system to a medium size of city in terms of collection, transport, processing and disposal (06 marks)

(Total 20 Marks)